From $x = \sin(-\theta)$, we must have $\sin^{-1} x = -\theta$, which in turn gives $\theta = -\sin^{-1} x$.

Hence, $\sin^{-1}(-x) = -\sin^{-1} x$.

(iv) If $x \in [-1, 1]$, then $-x \in [-1, 1]$. Thus, $\cos^{-1}(-x)$ is well defined

Let $\cos^{-1}(-x) = \theta$. Then $\theta \in [0, \pi]$ and $\cos \theta = -x$.

Now, $\cos\theta = -x$ implies $x = -\cos\theta = \cos(\pi - \theta)$.

Thus, $\pi - \theta = \cos^{-1} x$, which gives $\theta = \pi - \cos^{-1} x$.

Hence, $\cos^{-1}(-x) = \pi - \cos^{-1} x$.

Similarly, other results are proved.

Note

(i) The inverse function of an one-to-one and odd function is also an odd function. For instance, $y = \sin^{-1} x$ is an odd function, since sine function is both one-to-one and odd in the restricted domain $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

(ii) Is the inverse function of an even function also even? It turns out that the question does not make sense, because an even function cannot be one-to-one if it is defined anywhere other than 0. Observe that $\cos^{-1} x$ and $\sec^{-1} x$ are not even functions.

Property-V (Cofunction inverse identities)

(i) $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$, $x \in [-1, 1]$. (ii) $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$, $x \in \mathbb{R}$.

(iii) $\operatorname{cosec}^{-1} x + \sec^{-1} x = \frac{\pi}{2}$, $x \in \mathbb{R} \setminus (-1, 1)$ or $|x| \geq 1$.

Proof

(i) Here, $x \in [-1, 1]$. Let $\sin^{-1} x = \theta$. Then $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and $\sin\theta = x$.

Note that $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2} \Leftrightarrow 0 \leq \frac{\pi}{2} - \theta \leq \pi$.

So, $\cos\left(\frac{\pi}{2} - \theta\right) = \sin\theta = x$, which gives $\cos^{-1} x = \frac{\pi}{2} - \theta = \frac{\pi}{2} - \sin^{-1} x$.

Hence, $\cos^{-1} x + \sin^{-1} x = \frac{\pi}{2}$, $|x| \leq 1$.

(ii) Let $\cot^{-1} x = \theta$. Then, $\cot\theta = x$, $0 < \theta < \pi$ and $x \in \mathbb{R}$.

Now, $\tan\left(\frac{\pi}{2} - \theta\right) = \cot\theta = x$. ... (1)

Thus, for $x \in \mathbb{R}$, $\tan(\tan^{-1} x) = x$ and (1) gives $\tan(\tan^{-1} x) = \tan\left(\frac{\pi}{2} - \theta\right)$.

Hence, $\tan(\tan^{-1} x) = \tan\left(\frac{\pi}{2} - \cot^{-1} x\right)$ ... (2)

Note that $0 < \cot^{-1} x < \pi$ gives $-\frac{\pi}{2} < \frac{\pi}{2} - \cot^{-1} x < \frac{\pi}{2}$.

Thus, (2) gives $\tan^{-1} x = \frac{\pi}{2} - \cot^{-1} x$. So, $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$, $x \in \mathbb{R}$.

Similarly, (iii) can be proved.

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